CW Project: Quanser Helicopter

School of Science & Technology

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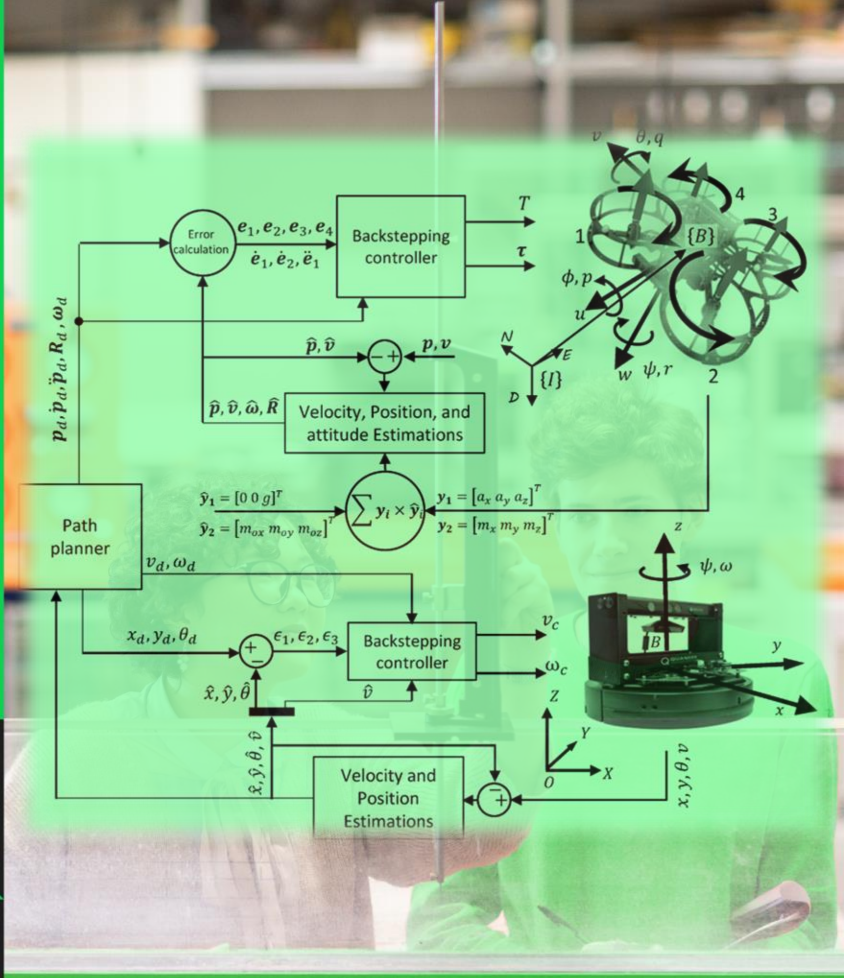

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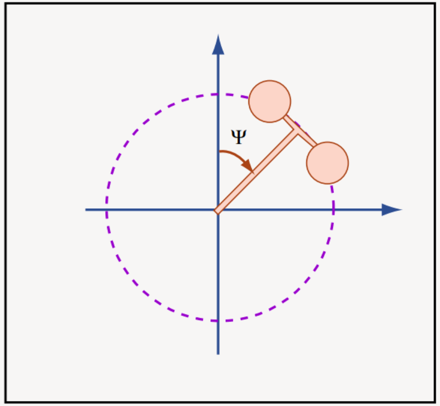
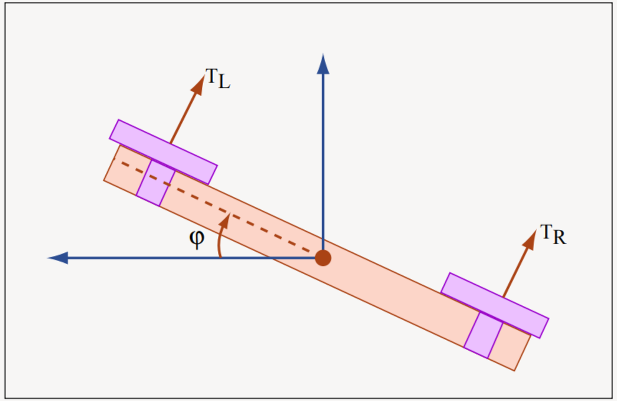
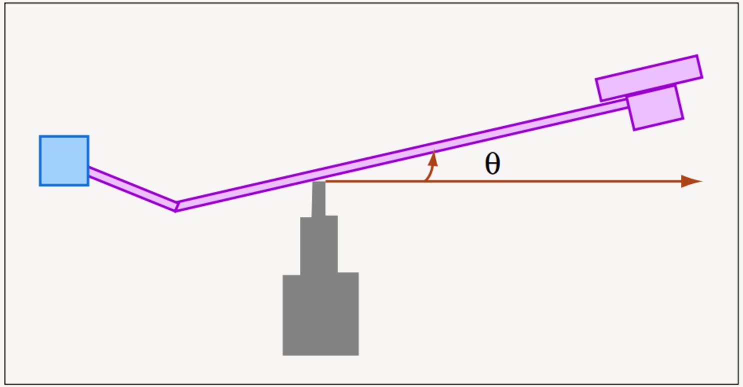
Part 1 (50%) : 1-Introduction

The Quanser helicopter is a mechanical device that emulates the flight of a reduced degree of freedom (DOF) helicopter. Instead of the usual six DOF of a free-flying helicopter, the Quanser only exhibits three DOF: roll, pitch, and travel, as illustrated in Figures 1-3. The Quanser system is actuated by two rotor speeds, and the inputs to the system are V_{cyc}, which is an electric voltage that results in differential change in the two rotor speeds, and V_{coll}, which is an electric voltage that controls the speed of the two propellers collectively. The outputs of the system are three angles: roll ϕ, pitch θ, and travel ψ.

Please note the limits of the Quanser: voltage $\in [-5, 5]$ Volts, $\phi \in [-40°, 40°]$, and $\theta \in [-25°, 30°]$.







Part 1: 2- Physical Model

The Quanser model is derived by applying Newton's second law to the rate of change of angular momentum. The nonlinear equations of motion are

$$\begin{aligned} I_{xx}\ddot{\phi} &= \tau_{cyc}l_h - mgl_\phi \sin(\phi) - L_p\dot{\phi} - I_r\omega_{rotor}(\dot{\theta}\cos(\phi) + \dot{\psi}\sin(\phi)), && (1) \\ I_{yy}\ddot{\theta} &= \tau_{coll}l_{boom}\cos(\phi) - Mgl_\theta \sin(\theta + \theta_{rest}) - Dl_{boom}\sin(\gamma) + I_r\omega_{rotor}\dot{\phi} - M_q\dot{\theta}, && (2) \\ I_{zz}\ddot{\psi} &= \tau_{coll}l_{boom}\sin(\phi) - Dl_{boom}\cos(\gamma), && (3) \end{aligned}$$

where D is the induced drag, τ_{coll} is the collective thrust, and τ_{cyc} is the cyclic thrust. These forces are modeled individually as

$$\begin{aligned} D &= K_D\dot{\psi}, && (4) \\ \tau_{coll} &= K_\tau\omega_{coll} - K_v\dot{\psi}, && (5) \\ \tau_{cyc} &= K_\tau\omega_{cyc} - K_v\dot{\psi}, && (6) \end{aligned}$$

where K_D, K_τ and K_v are constant coefficients. In addition, the motor dynamics can be modeled by

$$\begin{aligned} \dot{\omega}_{cyc} + 6\omega_{cyc} &= 780V_{cyc}, && (7) \\ \dot{\omega}_{coll} + 6\omega_{coll} &= 540V_{coll}. && (8) \end{aligned}$$

The values of other parameters in this model are given in Table 1. Note that ω_{rotor} is the rotor rotation speed, and is assumed to be constant; terms containing ω_{rotor} represent the rotors' contribution to angular momentum. Additionally, γ is the vehicle's flight path angle.

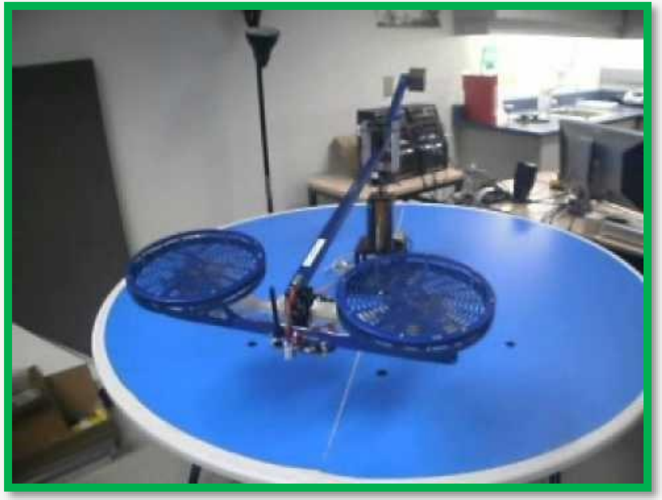
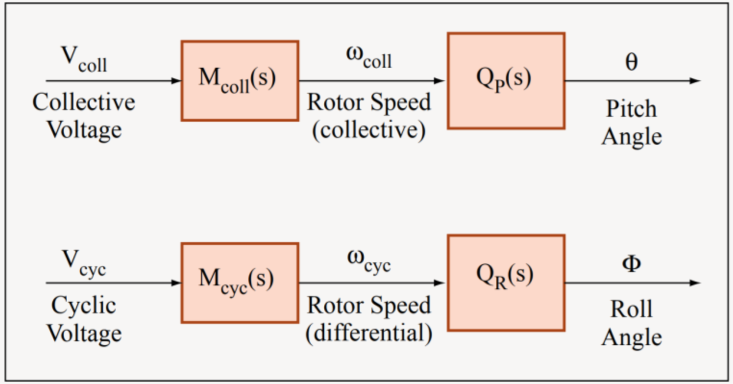


Table 1: Physical parameters

Parameter	Value	Units	Description
m	1.15	kg	mass of rotor assembly
M	3.57	kg	mass of whole setup
l_{boom}	0.66	m	length from pivot point to heli body
l_ϕ	0.004	m	length of pendulum for roll axis
l_θ	0.014	m	length of pendulum for pitch axis
l_h	0.177	m	length from pivot point to the rotor
I_{xx}	0.036	Nm	moment of inertia about x-axis
I_{yy}	0.93	Nm	moment of inertia about y-axis
K_τ	4.25×10^{-3}	N/s	coefficient of thrust
θ_{rest}	-25	degs	theta rest value
g	9.81	m/s^2	acceleration due to gravity
L_p	$[0.02, 0.2]$		roll damping coefficient
M_q	$[0.1, 0.9]$		pitch damping coefficient

Part 1: 2- Physical Model

Linearize these equations around a steady-state hover operating condition, i.e., $\dot{\phi}_0 = \dot{\theta}_0 = \dot{\psi}_0 = 0$, $\phi_0 = \theta_0 = \psi_0 = 0$, $\dot{\omega}_{cyc,0} = \dot{\omega}_{coll,0} = 0$, and $\gamma_0 = 0$. If we ignore the dynamics in travel (ψ), the pitch (θ) and roll (ϕ) subsystems can be represented by the block diagrams



The linear model does not perfectly represent the dynamics of the actual Quanser for many reasons. While there are many inaccuracies, the variations in pitch and roll damping between Quansers present the largest problem. You should assume that the damping coefficients L_p and M_q for any given Quanser will be somewhere in the ranges listed in Table 1.

Part 1: 3-Assignment

1. Derive the linearized model as described in Section 2, and symbolically form the transfer functions $M_{coll}(s), Q_P(s), M_{cyc}(s)$, and $Q_R(s)$. When forming the transfer functions, you may assume that the rotors' contribution to angular momentum is negligible (i.e. $\omega_{rotor} = 0$).

2. Use classical design techniques to develop controllers for the roll (ϕ) and pitch (θ) axes. You are encouraged to use Matlab/`sisotool` to assist you in your design. Remember that you are designing the controllers to work for the entire range of damping constants L_p and M_q. For the nominal plants (i.e., use a damping value in the middle of the range), your controllers must have the following properties:

   - Less than 10% error in tracking signals with frequency content up to 1 rad/s
   - At least 20° of phase margin
   - Maintain stability even for the worst-case damping

   For these controllers, write the controller transfer function and plot the following:

   - Bode plot of the controller
   - Open loop Bode plot of controller + nominal plant
   - Simulated closed-loop step response for a 20° step command
   - A paragraph or two describing your design strategy, predicted performance, and pros/cons of your design

Hint: Your roll (φ) controller is going to be the "inner-loop" for the travel controller in the next part, i.e. the travel controller will be commanding roll angle. Thus it is important that your roll controller is fast and stable but not so important that it has zero steady-state error. Conversely, your pitch (θ) controller should have zero steady state error so that you are sure to clear altitude obstacles in the next Part

Part 2 (50%): Designing a Travel Controller

In many aerospace systems, inner control loops are used to control the attitude of the plant, while outer loops are then used to control the location of the plant in space by commanding certain attitudes. Take, for instance, the altitude controller on most aircraft autopilots. The altitude controller itself commands a certain pitch, then an inner controller commands the elevator deflection to attain that pitch.

This is the strategy you will use to control the travel axis of the Quanser. Your travel controller will set the roll angle command, and that command goes through your roll angle controller, then is fed into the actual Quanser. This implies that your roll controller is now part of the system dynamics, and this fact must be accounted for in the system model.

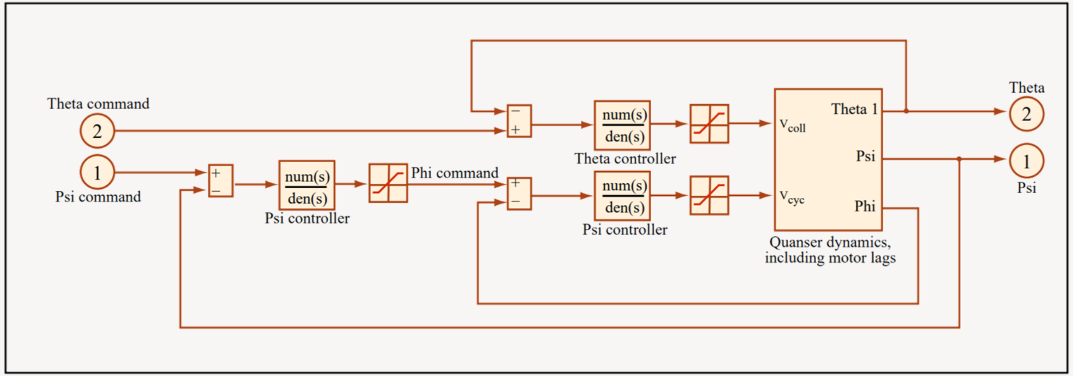
The overall architecture is shown in Figure 1. The model we want to derive is the transfer function from roll angle command to travel angle output, i.e.,

$$G^{\text{travel}}(s) = \frac{\psi(s)}{\phi_c(s)}.$$

The roll and pitch controllers provide the inner control loops:

$$\begin{aligned} V_{\text{coll}} &= G_c^{\text{pitch}}(s)(\theta_c - \theta), \\ V_{\text{cyc}} &= G_c^{\text{roll}}(s)(\phi_c - \phi). \end{aligned}$$

Figure 1: Overall architecture, all voltages must be saturated at ±5 V



The dynamics for the travel control design are established by closing these 2 inner loop controllers, then determining the dynamics from the input ϕ_c to the output ψ. The travel controller is then implemented as

$$\phi_c = G_c^{\text{travel}}(s)(\psi_c - \psi).$$

Table 1: Additional physical parameters (addendum to Lab 1)

Parameter	Value	Units	Description
I_{zz}	0.93	Nm	moment of inertia about z-axis
K_D	0		travel rate drag coefficient
$\bar{\omega}_{\text{coll}}$	$\frac{Mgl_\theta \sin(\theta_{rest})}{K_\tau l_{boom}}$	N	trim collective thrust
K_v	$0.0125\bar{\omega}_{\text{coll}} K_\tau l_{boom}$		travel rate thrust coefficient

Part 2: Assignment

1. **Augmented Plant Dynamics.** Select your preferred classical inner-loop controllers (e.g., from your own designs in Lab 1, or otherwise). Augment the linearized plant dynamics from Lab 1 with these controllers, as shown in Figure 1, and develop the linearized model for the travel plant $G^{\text{travel}}(s)$. Show explicitly how you developed this model.

2. **Classical Travel Controller.** Design a candidate, classical travel controller for the system. Please include:

   - A Bode plot of the controller
   - The open loop Bode plot of the controller + plant
   - Simulated step response for a 20° step command in travel angle
   - A paragraph or two describing your design strategy, predicted performance, and pros/cons of your design

Part 2: Assignment

3. **DOFB Pitch Controller.** Use state-space techniques to develop a candidate dynamic output feedback (DOFB) controller for pitch, $G_c^{\text{pitch}}(s)$. Your regulator should be designed via LQR, while your estimator should be designed via LQE Please include:

- A Bode plot of the controller
- The open loop Bode plot of the controller + plant
- Simulated step response for a 15° step command in pitch angle (include the input saturations as discussed in class)
- Simulate the effects of small sensor noise on your controllers to predict their flight test performance.
- A paragraph or two describing your design strategy, predicted performance, and pros/cons of your design
- Compare the DOFB and classical controllers, provide classical interpretations and discussing any key differences that you see.

Part 3: Bonus (10%)

This part will be a mission during which your Quanser must navigate over buildings, through a “tunnel,” and carry a payload back to the starting point. Penalties will be assessed for hitting the obstacles, which are virtual, and for the integral of the absolute error in both the pitch and travel axes. There will also be a penalty for not completing the “payload pickup” task properly (see the figure next slid). The team to perform the mission with the lowest score will win the bonus.

Mission

Figure 1 shows a map of the mission. The x-axis corresponds to the travel angle (ψ), while the y-axis corresponds to the pitch angle (θ). The buildings and tunnel will be physical obstacles that will fit within the constraints on the map (no tolerance information is given, however).

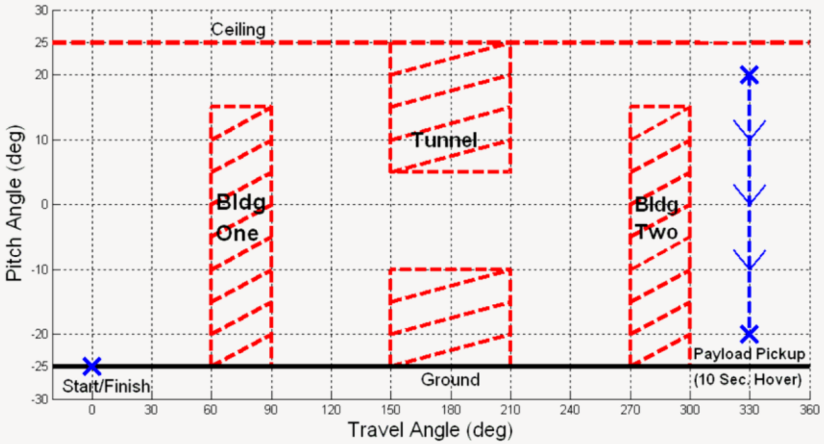


Figure 1: Mission map.

Part 3: Assignment

The mission has 3 stages:

1. Navigate from the starting point ($\psi = 0°$, $\theta = -25°$) to the surveillance waypoint ($\psi = 330°$, $\theta = 20°$), avoiding obstacles and staying below the ceiling ($\theta = 25°$).
2. Descend to the payload pickup waypoint ($\psi = 330°$, $\theta = -20°$), and hover there for 10 seconds. During this hover, a small payload will be attached to your Quanser.
3. Navigate back to the starting point ($\psi = 0°$, $\theta = -25°$), again avoiding obstacles and staying below the ceiling ($\theta = 25°$).

If you do not hover for a full 10 seconds at the payload pickup waypoint, 10 seconds will be added to your time for every second of hover missed.

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